

Chapter 10: Vectors

A Level Pure Maths Notes · Rexponential

Why this chapter matters

Vectors for describing motion — displacement, velocity, and force as quantities with both size and direction — won't be a new idea. A level Pure Maths formalises this: vectors as a tool for coordinate geometry and geometric proof (not just mechanics), extends everything into three dimensions, and revisits the mechanics applications with the calculus tools from Chapters 7–8 now available. This final chapter draws together ideas from across the whole course — coordinate geometry (Chapter 3), differentiation (Chapter 7) — so it's a fitting place to end.

Throughout, remember the core distinction: a vector has both magnitude (size) and direction, while a scalar has only magnitude. Displacement, velocity, force, and position (relative to an origin) are vectors; distance, speed, mass, and time are scalars.

10.1 Vectors in 2D: magnitude, direction, unit vectors

Concept

A 2D vector can be written in column form (x above y) or using the unit vectors i and j (i being a vector of length 1 in the x -direction, j likewise in the y -direction) — both notations mean exactly the same thing, and you should be comfortable converting between them. The magnitude of a vector is its length, found using Pythagoras (since the x - and y -components form a right-angled triangle with the vector as hypotenuse). A **unit vector** is any vector with magnitude exactly 1, and dividing any vector by its own magnitude always produces a unit vector pointing in the same direction — useful whenever a problem needs "just the direction" of something.

Procedure

For a vector $v = xi + yj$ (equivalently the column vector (x, y)):

- Magnitude: $|v| = \sqrt{x^2 + y^2}$
- Unit vector in the direction of v : $\hat{v}$ ("v-hat") $= v / |v| = (x/|v|)i + (y/|v|)j$
- Vector addition/subtraction and scalar multiplication work component-wise: $(x_1i + y_1j) + (x_2i + y_2j) = (x_1 + x_2)i + (y_1 + y_2)j$, and $k(xi + yj) = kxi + kyj$.

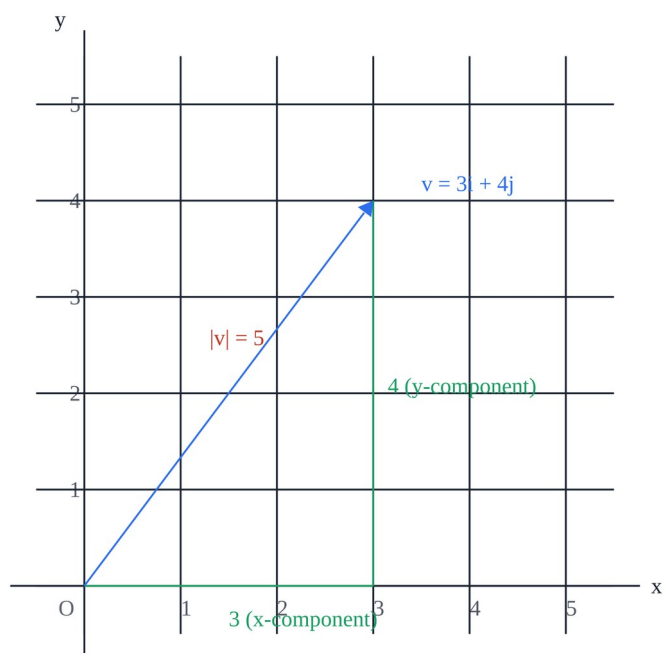
Worked example

EXAMPLE 10.1

Given the vector $v = 3i + 4j$, find (a) $|v|$, (b) the unit vector in the direction of v .

Step-by-step solution. (a) $|v| = \sqrt{3^2 + 4^2} = \sqrt{9+16} = \sqrt{25} = 5$ (M1, A1).

(b) $\hat{v} = v/|v| = (3i+4j)/5 = \frac{3}{5}i + \frac{4}{5}j$ (M1, A1). (Check: $|\hat{v}| = \sqrt{((3/5)^2 + (4/5)^2)} = \sqrt{(9/25 + 16/25)} = \sqrt{(25/25)} = 1$ ✓, confirming it's genuinely a unit vector.)



$v = 3i + 4j$, with its x - and y -components forming a right triangle

Common mistake

Forgetting to divide BOTH components by the magnitude when finding a unit vector — both the i - and j -components must be scaled by the same factor $1/|v|$; scaling only one component would change the vector's direction, not just its length.

10.2 Position vectors and dividing a line in a given ratio

Concept

A **position vector** describes the location of a point relative to a fixed origin O — the position vector of point A , written a (or OA with an arrow), is just the vector from O to A . This lets you do geometry using vector algebra: the vector from one point to another, AB , is always (position vector of B) minus (position vector of A), i.e. $AB = b - a$ — this single relationship underlies almost everything in this chapter.

Procedure

- Vector between two points: $AB = b - a$, where a, b are the position vectors of A, B .
- A point P dividing the line AB in the ratio $m:n$ (measured from A) has position vector: $p = a + \frac{m}{m+n} \times (b - a)$ — this comes from "start at A , and travel the appropriate fraction of the way along vector AB ." The midpoint (ratio $1:1$) is the special case $p = a + \frac{1}{2}(b-a) = \frac{1}{2}(a+b)$.

Worked example

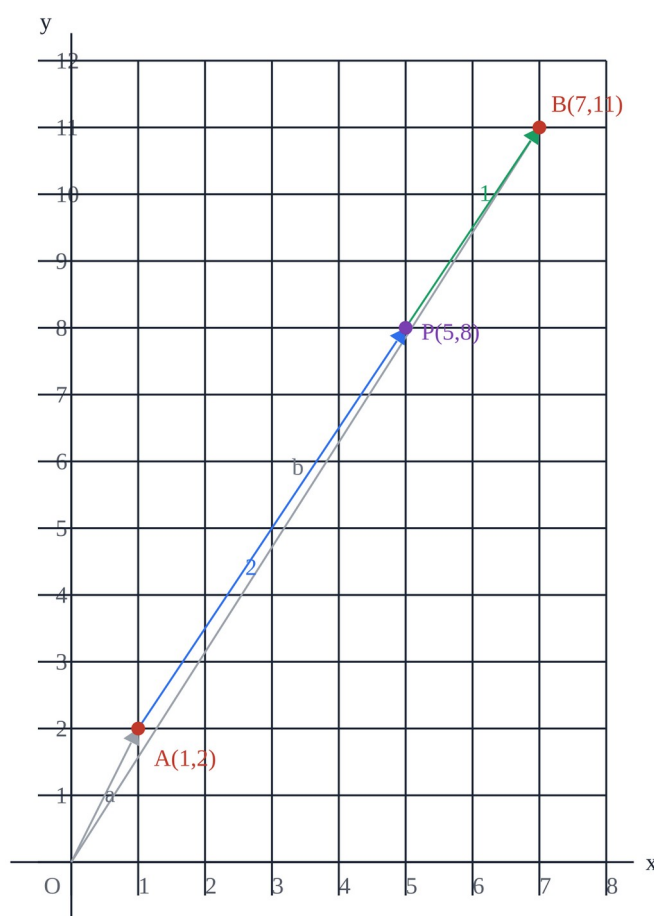
EXAMPLE 10.2

Points A and B have position vectors $\mathbf{a} = \mathbf{i} + 2\mathbf{j}$ and $\mathbf{b} = 7\mathbf{i} + 11\mathbf{j}$. Find the position vector of the point P that divides AB in the ratio 2:1 (from A to B).

Step 1 $\mathbf{AB} = \mathbf{b} - \mathbf{a} = (7\mathbf{i} + 11\mathbf{j}) - (\mathbf{i} + 2\mathbf{j}) = 6\mathbf{i} + 9\mathbf{j}$ (M1).

Step 2 Since P divides AB in the ratio 2:1, P is $2/(2+1) = 2/3$ of the way from A to B (M1).

Answer $\mathbf{p} = \mathbf{a} + \frac{2}{3}(\mathbf{AB}) = (\mathbf{i} + 2\mathbf{j}) + \frac{2}{3}(6\mathbf{i} + 9\mathbf{j}) = (\mathbf{i} + 2\mathbf{j}) + (4\mathbf{i} + 6\mathbf{j}) = \mathbf{5i} + \mathbf{8j}$ (A1). (Check: the point (5,8) should lie on the straight line through (1,2) and (7,11) — gradient from A to P is $(8-2)/(5-1) = 6/4 = 1.5$, and from A to B is $(11-2)/(7-1) = 9/6 = 1.5$ — same gradient, confirming P lies on line AB ✓.)



P divides AB in ratio 2:1 from A; $\mathbf{OP} = \mathbf{OA} + (2/3)\mathbf{AB}$

Common mistake

Using the ratio the wrong way round — for a ratio $m:n$ from A to B, the fraction of the way from A is $m/(m+n)$, NOT $n/(m+n)$; if the answer doesn't lie sensibly between A and B when you check it, this is the first thing to recheck.

10.3 Vector geometry: collinearity, parallelism, and shape properties

Concept

Vectors give an elegant, purely algebraic way to prove geometric facts — no need for a carefully-drawn diagram or a synthetic (angle-chasing) argument. Two of the most common proof types: **collinearity** (three points lying on a single straight line) and **parallelism** (two line segments pointing in the same, or exactly opposite, direction). Both come down to the same underlying test: is one vector a scalar multiple of another?

Procedure

- Two vectors u and v are **parallel** if and only if $u = k \cdot v$ for some scalar k (they point along the same line, though possibly opposite directions if k is negative).
- To prove three points A, B, C are **collinear**: show that AB and BC (or AB and AC) are parallel (one is a scalar multiple of the other) AND that they share a common point (which they do automatically, since both vectors start or pass through a shared point like B) — in practice, showing $AB = k \cdot BC$ for some scalar k , with B common to both, is sufficient.
- These techniques extend naturally to proving quadrilaterals are parallelograms (opposite sides are parallel and equal, i.e. represented by the same vector), showing points bisect each other, and so on.

Worked example

EXAMPLE 10.3

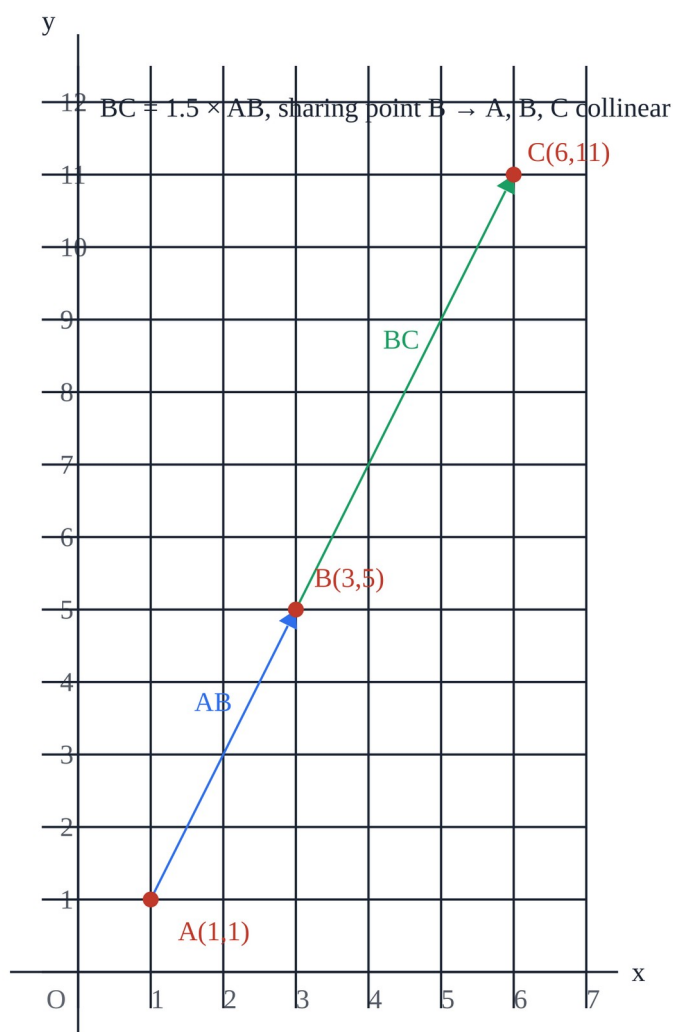
Points A, B, C have position vectors $a = i + j$, $b = 3i + 5j$, $c = 6i + 11j$. Prove that A, B , and C are collinear.

Step 1 $AB = b - a = (3i+5j) - (i+j) = 2i + 4j$ (M1).

Step 2 $BC = c - b = (6i+11j) - (3i+5j) = 3i + 6j$ (M1).

Step 3 Compare: is $3i+6j$ a multiple of $2i+4j$? $3/2 = 1.5$ and $6/4 = 1.5$ — yes, both components scale by 1.5, so **$BC = 1.5 \times AB$** (A1).

Answer Since BC is a scalar multiple of AB , the vectors AB and BC are parallel; since they also share the common point B , the points A, B , and C must lie on the same straight line — **A, B, C are collinear** (A1, with the conclusion clearly stated, not just the scalar relationship).



A, B, C collinear: AB and BC are parallel and share point B

Common mistake

Showing two vectors are parallel but forgetting to explicitly state the "shared point" part of the argument — two parallel vectors that DON'T share a common point describe two separate, distinct parallel lines, not one straight line through three points; the conclusion of collinearity needs both parts of the argument stated clearly.

10.4 Vectors in 3D

Concept

Everything from 10.1–10.3 extends naturally into three dimensions by simply adding a third component (a k -direction, perpendicular to both i and j) — the algebra works exactly the same way, just with one more coordinate to track. The main genuinely new tool at this level is the **scalar (dot) product**, which combines two vectors to produce a single number, and is closely connected to the angle between them.

Procedure

For 3D vectors $u = (u_1, u_2, u_3)$ and $v = (v_1, v_2, v_3)$:

- Magnitude: $|u| = \sqrt{u_1^2 + u_2^2 + u_3^2}$
- Addition, subtraction, scalar multiplication: component-wise, exactly as in 2D.
- **Dot product:** $u \cdot v = u_1v_1 + u_2v_2 + u_3v_3$ (a scalar, not a vector — note this is different from adding or multiplying component-wise into a new vector).
- The dot product connects to the angle θ between the vectors via: $u \cdot v = |u||v|\cos\theta$, which rearranges to give the angle: $\cos\theta = (u \cdot v) / (|u||v|)$.
- Two vectors are **perpendicular** if and only if their dot product is zero (since $\cos 90^\circ = 0$) — a very quick test worth remembering.

Worked example

EXAMPLE 10.4

Given $a = (1, 2, 3)$ and $b = (2, -1, 4)$, find (a) $a \cdot b$, (b) the angle between a and b (to 1 d.p.).

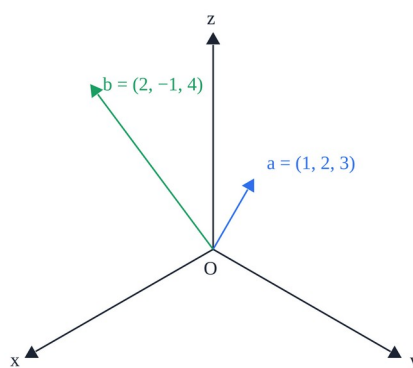
Step 1 (a) $a \cdot b = (1)(2) + (2)(-1) + (3)(4) = 2 - 2 + 12 = 12$ (M1, A1).

Step 2 (b) $|a| = \sqrt{1+4+9} = \sqrt{14}$. $|b| = \sqrt{4+1+16} = \sqrt{21}$ (M1).

Step 3 $\cos\theta = 12/(\sqrt{14} \times \sqrt{21}) = 12/\sqrt{294} = 12/17.146... = 0.6998...$ (M1).

Answer $\theta = \cos^{-1}(0.6998) = 45.6^\circ$ (1 d.p.) (A1).

a and b sketched in 3D (isometric view)



$$a \cdot b = (1)(2) + (2)(-1) + (3)(4) = 12$$

$$\theta = \cos^{-1}(12 / (\sqrt{14} \times \sqrt{21})) \approx 45.6^\circ$$

a and b sketched in 3D, with the angle between them found from $a \cdot b$

Common mistake

Confusing the dot product (a scalar, from $u_1v_1 + u_2v_2 + u_3v_3$) with simple component-wise multiplication into a new vector (u_1v_1, u_2v_2, u_3v_3) — the dot product specifically SUMS these products into one number; if your "dot product" answer has three components, you've computed something else entirely.

10.5 Vectors in mechanics

Concept

This section closes the loop between vectors and calculus (Chapter 7): when a particle's position is described by a vector that changes with time, differentiating that position vector component-wise gives the velocity vector, and differentiating again gives the acceleration vector — exactly the same relationship as in 1D kinematics, just applied to each component of the vector independently.

Procedure

For a position vector $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$ (components depending on time t):

- Velocity: $\mathbf{v}(t) = d\mathbf{r}/dt = (dx/dt)\mathbf{i} + (dy/dt)\mathbf{j}$ — differentiate each component separately.
- Acceleration: $\mathbf{a}(t) = d\mathbf{v}/dt = d^2\mathbf{r}/dt^2$ — differentiate again, component-wise.
- **Speed** is the magnitude of the velocity vector: $\text{speed} = |\mathbf{v}(t)| = \sqrt{[(dx/dt)^2 + (dy/dt)^2]}$ (a scalar, distinct from velocity itself, which is a vector).

Worked example

EXAMPLE 10.5

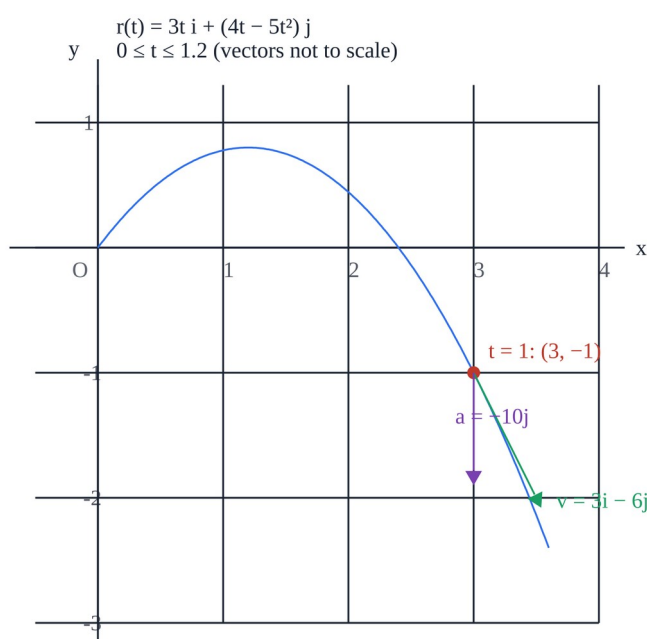
A particle's position vector at time t seconds is $\mathbf{r}(t) = 3t\mathbf{i} + (4t - 5t^2)\mathbf{j}$ (metres). Find (a) the velocity vector, (b) the acceleration vector, (c) the speed of the particle at $t = 1$ second.

Step 1 (a) Differentiate each component with respect to t : $\mathbf{v}(t) = d\mathbf{r}/dt = 3\mathbf{i} + (4 - 10t)\mathbf{j}$ (M1, A1).

Step 2 (b) Differentiate again: $\mathbf{a}(t) = d\mathbf{v}/dt = -10\mathbf{j}$ (M1, A1 — note the $\mathbf{i}$ -component's derivative is 0, since the x -component $3t$ has a constant rate of change, so there's no acceleration in the x -direction; this makes physical sense, as this models a projectile with constant horizontal velocity and constant downward acceleration, like gravity).

Step 3 (c) At $t=1$: $\mathbf{v}(1) = 3\mathbf{i} + (4-10)\mathbf{j} = 3\mathbf{i} - 6\mathbf{j}$ (M1).

Answer Speed = $|\mathbf{v}(1)| = \sqrt{(3^2 + (-6)^2)} = \sqrt{(9+36)} = \sqrt{45} = 3\sqrt{5} \approx 6.71 \text{ m/s}$ (3 s.f.) (A1).



Trajectory $r(t)$, with velocity and acceleration vectors at $t = 1$

Common mistake

Reporting the velocity vector $(3, -6)$ when asked for **speed** — speed is a single non-negative number (the magnitude of velocity), not a vector; always check whether a question wants the vector quantity or its magnitude, and give exactly the one asked for.

Chapter summary

Key formulas

KEY FORMULAS

Magnitude

$$|v| = \sqrt{x^2 + y^2} \quad (2D) \quad |v| = \sqrt{x^2 + y^2 + z^2} \quad (3D)$$

Unit vector

$$\hat{v} = v / |v|$$

Vector between two points

$$AB = b - a$$

Point dividing AB in ratio $m:n$ from A

$$p = a + [m/(m+n)](b - a)$$

Parallel vectors

$$u = k \cdot v, \text{ for some scalar } k$$

Dot product

$$u \cdot v = u_1v_1 + u_2v_2 + u_3v_3 = |u||v|\cos\theta \quad \text{perpendicular} \iff u \cdot v = 0$$

Mechanics

$$v(t) = dr/dt, \quad a(t) = dv/dt, \quad \text{speed} = |v(t)|$$

Common exam mistakes across this chapter

- Only scaling one component when finding a unit vector.
- Using the wrong fraction ($m/(m+n)$ vs $n/(m+n)$) when dividing a line in a given ratio.
- Proving vectors parallel but forgetting to also state the shared-point condition needed for collinearity.
- Computing the dot product as a vector (component-wise products) instead of summing them into a scalar.
- Confusing velocity (a vector) with speed (its magnitude, a scalar) in mechanics questions.

You should now be able to

- Work with 2D and 3D vectors: magnitude, unit vectors, position vectors, and dividing a line in a given ratio
- Use vectors to prove collinearity, parallelism and other geometric properties
- Use the scalar (dot) product to find the angle between two vectors and test for perpendicularity
- Differentiate a position vector to find velocity, acceleration and speed

Looking back. This completes all ten chapters of the A Level Pure Maths notes — from the logical foundations of proof, through algebra, coordinate geometry, sequences and series, trigonometry, exponentials and logarithms, the full arc of differentiation and integration, numerical methods, and finally vectors. Chapters 7 (Differentiation), 8 (Integration), and 9 (Numerical Methods) cover the most substantial new ground and are worth the most revisiting as you prepare for your exams.